

Solution: Vision Desk

ViT, JSON and grounded MCP

Computer Vision in Finance — Master's in Finance, Data & AI

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1 How to read this solution

This document answers the case **closed task** with a reference protocol and sample traces. Section 2.1 records metrics from an executed reference notebook (**SEED=42**). Other F1 values remain pedagogical. Non-negotiable: ViT (not LSTM), JSON schema before test, four MCP tools, and no invented claims.

2 Reference lab (Colab)

Runnable notebook aligned with this solution:

Resource	Link
Repository	github.com/LikhitaYerra/vision-desk
Notebook	vision_desk_colab.ipynb
Open in Colab	colab.research.google.com/.../vision_desk_colab.ipynb

The notebook implements: synthetic finance images, split by document, ViT head fine-tune vs majority baseline, JSON contract (`vision.extract`), four MCP stubs, plus grounded and refusal traces. Reference outputs live in **lab-reference/** (see below). Groups may replace synthetic data with their own captures if the protocol stays the same.

2.1 Recorded run (executed notebook, SEED=42)

The Colab notebook code was executed end-to-end (local CPU, October 2026; same logic and seed as [Colab](#)). Artifacts are copied in the repo under **lab-reference/** (`metrics.json`, `grounded_trace.json`, `refusal_trace.json`).

Metric	Value (held-out documents)
Train / val images	96 / 32 (8 val documents)
Majority baseline macro-F1	0.056
ViT-Tiny head fine-tune macro-F1	1.000 (8 epochs)
Beats baseline?	yes
Saved files	<code>metrics.json</code> , <code>grounded_trace.json</code> , <code>refusal_trace.json</code>

Reading. Macro-F1 = 1.0 on the *synthetic* templates is expected: layouts are clean and classes are visually distinct. Real Bloomberg / PDF captures should show lower F1 and a richer confusion matrix — an honest negative result is fine if the split-by-document protocol and MCP traces are correct.

Sample `vision.extract` (**val**, **with ticker**).

```
{
  "type": "candlestick_chart",
  "confidence": 0.989,
  "ticker": "AAPL",
  "fields": {"title": "candlestick_chart__doc05_p0", "period": "2024"},
  "source_tool": "vision.extract"
}
```

Grounded answer (from `lab-reference/grounded_trace.json`). “`vision.classify` labels this as `candlestick_chart` (score 0.989). `vision.extract` returns `ticker=AAPL`. `market.lookup`: Apple Inc. (USD). I assert no closing price: no price field is present in the JSON.”

Refusal (from `lab-reference/refusal_trace.json`). “I cannot confirm a ticker: extraction returned null or `market.lookup` failed. I refuse to invent a symbol. Retry with a wider crop / different DPI.”

3 Answer to the closed task (structure)

Requirement	What a good group shows
1. Vision assertion	Class + score from <code>vision.classify</code> ; fields from <code>vision.extract</code>
2. Tool justification	Each sentence cites a tool call and a JSON field
3. Refusal	If ticker is null / tool KO → refuse or retry, never invent

4 Chosen protocol

Choice	Value
Input	Page / chart captures, 2–4 classes
Split	By document (lab: 75 % / 25 % held-out docs; 96 / 32 images)
Model	Pretrained ViT-Tiny (or Small) + fine-tuned head
Baseline	Majority class, or logits on reduced pixels (session 1)
JSON	<code>type</code> , <code>confidence</code> , <code>ticker?</code> (nullable), <code>fields</code>
MCP	<code>vision.classify</code> , <code>vision.extract</code> , <code>market.lookup</code> , <code>knowledge.search</code>
LLM	Asserts a fact only if it appears in a tool JSON

5 JSON schema (contract)

```
{
  "type": "candlestick_chart",
  "confidence": 0.989,
  "ticker": "AAPL",
  "fields": {
    "title": "candlestick_chart__doc05_p0",
    "period": "2024"
  },
}
```

```
"source_tool": "vision.extract"
}
```

Same payload as the recorded lab sample above (title = image stem). `ticker` may be `null`. Invalid JSON (missing required field) is a pipeline failure even if the class is correct.

6 Vision results (expected order of magnitude)

Model	Macro F1 (held-out)	Reading
Reference lab (synthetic)	maj. 0.06 / ViT 1.00	See §2.1; not a prod. claim
Majority baseline (real data)	≈ 0.25 –0.40	Depends on imbalance
Fine-tuned ViT	clearly higher	If layout is stable
ViT on new template	often lower	Expected domain shift

Reading. If held-out mixes pages from the same PDF as train, F1 is inflated (document leakage). Analogue of random `train_test_split` in Signal Desk.

7 Grounded trace (requirement 1 + 2)

User question. “What type of image is this, and which ticker appears?”

Tools.

1. `vision.classify` → {class: `candlestick_chart`, score: 0.989}
2. `vision.extract` → JSON above (`ticker`: `"AAPL"`)
3. `market.lookup("AAPL")` → metadata (name, currency) — stub OK
4. `knowledge.search("candlestick")` → short glossary — stub OK

Acceptable final answer. “`vision.classify` labels the image as a candlestick chart (0.989). `vision.extract` returns `ticker=AAPL`. `market.lookup` confirms Apple Inc. metadata. I assert no closing price: no price field is present in the JSON.”

8 Refusal after tool failure (requirement 3)

Situation. `vision.extract` returns `ticker: null` (or `market.lookup` returns an error).

Acceptable. “I cannot confirm a ticker: extraction returns null. I refuse to invent a symbol. Retry with a wider crop / different DPI.”

Unacceptable. “It’s clearly Tesla.” — no tool source.

9 Penalized anti-patterns

- LSTM or Conv1D on the image “because deep learning”;
- LLM claim without a tool call;
- calling `vision.*` then ignoring the JSON;
- presenting a RAG glossary as OCR / extraction;
- write / mutate tools inside the lab (read-only MCP only).

10 Typical limits (copy into the 4-page note)

1. **Layout shift** — new template $\Rightarrow$ F1 drop.
2. **Capture** — DPI, UI chrome, JPEG compression.
3. **Hallucination** — the LLM fills what the JSON does not have.

11 Link with Signal Desk

Signal Desk	Vision Desk
OHLCV series + LSTM	Image + ViT
Naïve baseline / costs	Class baseline + MCP grounding
Walk-forward	Split by document + held-out templates
Dead signal after costs = OK	Refusal after tool KO = OK

12 Video resources

- [3Blue1Brown — Attention](#);
- [Kilcher — ViT paper](#);
- [Vision Transformer in PyTorch](#);
- [MCP — Anthropic](#);
- [MCP beginners Python](#).